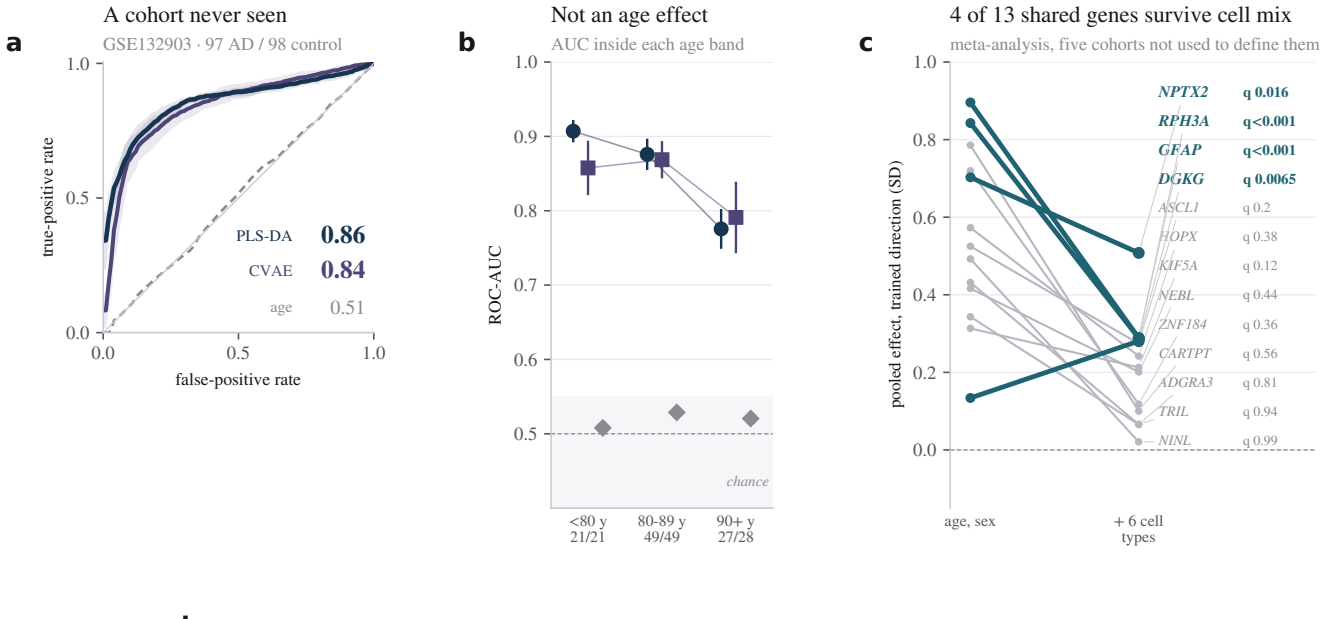

d Evidence map: frozen gene signatures in every validation cohort, no model refitted

	Cohort	AD/ctrl	Separates AD from control	Beats matched random genes	Beats plain top-N DE	Adds beyond cell mix	Tracks Braak stage	Tracks plaques or MMSE
held out	**GSE132903** Illumina HT-12 v4 · middle temporal gyrus	97/98	0.89 0.85	p<.001 p<.001	+0.04 +0.02	+0.11 +0.11	—	—
independent microarray	**GSE15222** Illumina RefSeq-8 · temporal cortex	176/187	0.84 0.79	p<.001 p .10	+0.02 -0.02	+0.04 +0.01	—	—
independent microarray	**GSE29378** Illumina HT-12 v3 · hippocampus (CA1, CA3)	17/16	0.88 0.70	p<.001 p .51	+0.14 -0.04	+0.13 +0.01	+0.46 -0.06	+0.50 -0.08
independent microarray	**GSE122063** Agilent 8x60K · frontal + temporal cortex	12/11	0.90 0.89	p .01 p .09	+0.01 0.00	+0.02 +0.01	—	—
independent microarray	**GSE36980** Affymetrix HuGene 1.0 ST · frontal cortex	15/17	0.75 0.84	p .10 p<.001	-0.02 +0.08	+0.03 0.00	—	—
RNA-seq	**GSE125583** RNA-seq (HiSeq 2500) · fusiform gyrus	219/70	0.83 0.82	p<.001 p .01	+0.02 0.00	+0.06 +0.06	+0.26 +0.35	—
pathology only	**GSE106241** Agilent 8x60K custom · inferior temporal cortex	*58 graded*	—	—	—	—	+0.44 +0.49	—
pathology only	**GSE1297** Affymetrix U133A · hippocampus	*31 graded*	—	—	—	—	+0.17 0.00	+0.42 +0.01
	tests passed (PLS-DA | CVAE)			**5/6 3/6**	**5/6 3/6**	**5/6 2/6**	**2/4 2/4**	**1/2 0/2**

left half PLS-DA, right half CVAE · solid = test passed ($p < 0.05$, or above the comparison for plain DE) · pale = not passed
values below each disc: AUC, p against matched random genes, AUC minus plain DE, ΔAUC over cell mix, partial ρ with pathology